

Lista 3 de Geometria Analítica. Jeann Victor Batista.

① $AX=B$

$$\begin{Bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{Bmatrix} \rightarrow \begin{Bmatrix} 1 & 3 & 4 \\ -1 & 2 & 5 \\ -2 & -1 & 3 \end{Bmatrix} \text{ De acordo as condições impostas}$$

$$\begin{Bmatrix} 1 & 3 & 4 \\ -1 & 2 & 5 \\ -2 & -1 & 3 \end{Bmatrix} \cdot \begin{Bmatrix} a \\ b \\ c \end{Bmatrix} \rightarrow \begin{Bmatrix} -1 \\ 2 \\ -3 \end{Bmatrix} \rightarrow \begin{cases} 1.a + 3.b + 4.c = -1 \\ -1.a + 2.b + 5.c = 2 \\ -2.a + -1.b + 3.c = -3 \end{cases}$$

$$\begin{aligned} a + 3b + 4c &= -1 \\ -a + 2b + 5c &= 2 \\ -2a - b + 3c &= -3 \end{aligned} \rightarrow \begin{aligned} a + 3b + 4c &= -1 && \text{Linha 2 + Linha 1} \\ 5b + 9c &= 1 && \text{Linha 3 + Linha 1} \\ 5b + 11c &= -5 && \end{aligned}$$

$$\begin{aligned} a + 3b + 4c &= -1 \\ 5b + 9c &= 1 \\ -2c &= 6 \end{aligned} \rightarrow \begin{aligned} -2c &= 6 \rightarrow c = 6/-2 \rightarrow c = -3 \\ 5b + 9c &= 1 \rightarrow 5b + (-27) = 1 \rightarrow 5b = 28 \rightarrow b = 5.6 \\ a + 3b + 4c &= -1 \rightarrow a + 16.8 + (-12) = -1 \rightarrow a = -9.8 \end{aligned}$$

$$X = \begin{bmatrix} -9.8 \\ 5.6 \\ -3 \end{bmatrix}$$

②

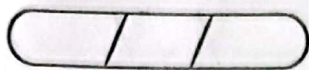
$$a) \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \cdot X = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \rightarrow A^{-1} \cdot \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \cdot X = A^{-1} \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot \text{adjunta} \rightarrow \det A = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} = 3 - 8 \rightarrow \det A = -5$$

$$\text{Matriz cofatora pelo método é: } \begin{pmatrix} 3 & -2 \\ -4 & 1 \end{pmatrix} // \text{Adjunta é: } \begin{pmatrix} 3 & -4 \\ -2 & 1 \end{pmatrix}$$

$$A^{-1} = -\frac{1}{5} \cdot \begin{pmatrix} 3 & -4 \\ -2 & 1 \end{pmatrix} \rightarrow A^{-1} = \begin{pmatrix} -3/5 & 4/5 \\ 2/5 & -1/5 \end{pmatrix}$$

$$\begin{pmatrix} -3/5 & 4/5 \\ 2/5 & -1/5 \end{pmatrix} \cdot \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \cdot X = \begin{pmatrix} -3/5 & 4/5 \\ 2/5 & -1/5 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} -3/5 & 4/5 \\ 2/5 & -1/5 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$



$$X = \begin{pmatrix} -315 & 115 \\ 215 & -115 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix} \rightarrow X = \begin{pmatrix} 315 + 115 \\ -215 + 115 \end{pmatrix} \rightarrow X = \begin{pmatrix} -115 \\ -115 \end{pmatrix}$$

b)

$$\begin{pmatrix} 2 & 3 \\ 5 & 5 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} \cdot Y = \begin{pmatrix} 1 & 7 \\ 2 & 7 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} \cdot Y = \begin{pmatrix} 1 & 7 \\ 2 & 7 \end{pmatrix} + \begin{pmatrix} -2 & -3 \\ -5 & -5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} \cdot Y = \begin{pmatrix} -1 & 4 \\ -3 & 2 \end{pmatrix} \quad \left\{ \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} \cdot 5 - 6 \rightarrow \begin{pmatrix} 5 & 3 \\ -2 & 1 \end{pmatrix} \rightarrow A^{-1} = \frac{1}{-1} \cdot \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix} \right.$$

$$A^{-1} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} \cdot Y = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 4 \\ -3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot Y = \begin{pmatrix} 5 & -6 & -20 & 4 \\ -3 & 3 & 12 & -2 \end{pmatrix} \quad Y = \begin{pmatrix} -1 & -16 \\ 0 & 10 \end{pmatrix}$$

c)

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix} \cdot W = \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \rightarrow \text{Pela propriedade de determinantes, matrizes triangulares inferiores, a determinante é = D principal multiplicada. Det = 1.}$$

Matriz Cop =

$$a_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 1 & 0 \\ 3 & 1 \end{vmatrix} \rightarrow 1 \cdot 1 = 1 \quad \left\{ a_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 2 & 0 \\ 2 & 1 \end{vmatrix} = -1 \cdot 2 = -2 \right.$$

$$a_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} \rightarrow 1 \cdot (2 \cdot 3 - 2 \cdot 2) = 1 \cdot (6 - 4) = 2 \quad \left\{ a_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 0 & 0 \\ 3 & 1 \end{vmatrix} = -1 \cdot 0 = 0 \right.$$

$$a_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 = 1 \quad \left\{ a_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 1 & 0 \\ 2 & 3 \end{vmatrix} = -1 \cdot 3 = -3 \right.$$

$$a_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} = 1 \cdot 0 = 0 \quad \left\{ a_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 1 & 0 \\ 2 & 0 \end{vmatrix} = -1 \cdot 0 = 0 \right. \quad \left\{ a_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 = 1 \right.$$

$$\begin{pmatrix} 1 & -2 & 4 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{+} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 4 & -3 & 1 \end{pmatrix} \xrightarrow{A^{-1} = \frac{1}{1}} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 4 & -3 & 1 \end{pmatrix} \rightarrow A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 4 & -3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 4 & -3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix} \cdot W = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 4 & -3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot W = \begin{pmatrix} 5 & 0 & 0 \\ -10 & 7 & 0 \\ 20 & -21 & 2 \end{pmatrix} \rightarrow W = \begin{pmatrix} 5 \\ -3 \\ 1 \end{pmatrix}$$

③ a) $AXB = C \rightarrow (A^{-1} \cdot A) \cdot XB = A^{-1} \cdot C$

$$I \cdot XB = A^{-1} \cdot C \rightarrow XB \cdot B^{-1} = A^{-1} \cdot C \cdot B^{-1}$$

$$X = A^{-1} \cdot C \cdot B^{-1}$$

b) $A(B+X) = A \rightarrow AB + AX = A \rightarrow AX = A - AB \quad A^{-1} \cdot AX = A^{-1} \cdot A - A^{-1} \cdot AB$

$$X = I_n - AB$$

c) $ACXB = C \rightarrow A^{-1} \cdot ACXB = A^{-1} \cdot C \rightarrow CXB = A^{-1} \cdot C$

$$C^{-1} \cdot CXB = C^{-1} \cdot A^{-1} \cdot C \rightarrow XB = A^{-1} \rightarrow XB \cdot B^{-1} = A^{-1} \cdot B^{-1} \rightarrow X = A^{-1} \cdot B^{-1}$$

d) $(AB)^{-1} \cdot (AX) = CC^{-1} \rightarrow (AB)^{-1} \cdot (AX) = I_n \rightarrow B^{-1} \cdot A^{-1} \cdot A \cdot X = I_n$

$$B^{-1} \cdot X = I_n \rightarrow B \cdot B^{-1} \cdot X = B \cdot I_n \rightarrow X = B$$

e) $AB^t X B^{-1} = A^t \rightarrow A^{-1} \cdot A \cdot B^t X B^{-1} = A^{-1} \cdot A^t \rightarrow B^t X B^{-1} = A^{-1} \cdot A^t$

$$B \cdot B^t X B^{-1} = A^{-1} \cdot B \cdot A^t \rightarrow X B^{-1} \cdot B = A^{-1} \cdot A^t \cdot B \cdot B^{-1} \rightarrow X = A^{-1} \cdot A^t \cdot B \cdot B^{-1}$$

$$a) \begin{cases} 3x - 2y = 1 \\ x + 3y = 9 \end{cases} \rightarrow \begin{Bmatrix} 3 & -2 \\ 1 & 3 \end{Bmatrix} \cdot \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} 1 \\ 9 \end{Bmatrix}$$

$$\det A = \begin{vmatrix} 3 & -2 \\ 1 & 3 \end{vmatrix} = 3 \cdot 3 - (-2) = 9 + 2 = 13 \quad \left\{ \begin{array}{l} \det A = \begin{vmatrix} 3 & 1 \\ 1 & 9 \end{vmatrix} = 3 \cdot 9 - 1 = 26 \end{array} \right.$$

$$\det X = \begin{vmatrix} 1 & -2 \\ 9 & 3 \end{vmatrix} = 3 \cdot 1 - (9 \cdot -2) = 3 + 18 = 21$$

$$X = \frac{\det X}{\det A} = \frac{21}{13} \quad Y = \frac{\det Y}{\det A} = \frac{26}{13} = 2$$

$$(X, Y) = (3, 2)$$

$$b) \begin{cases} 5x + 8y = 34 \\ 10x + 16y = 50 \end{cases} \rightarrow \begin{Bmatrix} 5 & 8 \\ 10 & 16 \end{Bmatrix} \cdot \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} 34 \\ 50 \end{Bmatrix}$$

$$\det A = \begin{vmatrix} 5 & 8 \\ 10 & 16 \end{vmatrix} = 5 \cdot 16 - 8 \cdot 10 = 80 - 80 = 0 \quad \left\{ \begin{array}{l} \det X = \begin{vmatrix} 34 & 8 \\ 50 & 16 \end{vmatrix} = 544 - 800 = -256 \end{array} \right.$$

$$\det Y = \begin{vmatrix} 5 & 34 \\ 10 & 50 \end{vmatrix} = 250 - 340 = -90$$

Este é um sistema impossível.

$$c) \begin{cases} x + 2y = 5 \\ 2x - 3y = -21 \end{cases} \rightarrow \begin{Bmatrix} 1 & 2 \\ 2 & -3 \end{Bmatrix} \cdot \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} 5 \\ -21 \end{Bmatrix}$$

$$\det A = \begin{vmatrix} 1 & 2 \\ 2 & -3 \end{vmatrix} = 1 \cdot (-3) - 2 \cdot 2 = -3 - 4 = -7 \quad \left\{ \begin{array}{l} \det X = \begin{vmatrix} 5 & 2 \\ -21 & -3 \end{vmatrix} = 5 \cdot (-3) - 2 \cdot (-21) = -15 + 42 = 27 \\ \det Y = \begin{vmatrix} 1 & 5 \\ 2 & -21 \end{vmatrix} = 1 \cdot (-21) - 5 \cdot 2 = -21 - 10 = -31 \end{array} \right.$$

$$\det Y = \begin{vmatrix} 1 & 5 \\ 2 & -21 \end{vmatrix} = 1 \cdot (-21) - 5 \cdot 2 = -21 - 10 = -31$$

$$X = \frac{\det X}{\det A} = \frac{27}{-7} = -\frac{27}{7} \quad Y = \frac{\det Y}{\det A} = \frac{-31}{-7} = \frac{31}{7}$$

$$d) \begin{cases} 3x + 2y - 5z = 8 \\ 2x - y - 2z = -1 \\ x - 2y - 3z = -1 \end{cases} \rightarrow \begin{matrix} x & y & z \\ \begin{pmatrix} 3 & 2 & -5 \\ 2 & -1 & -2 \\ 1 & -2 & -3 \end{pmatrix} & = & \begin{pmatrix} 8 \\ -1 \\ -1 \end{pmatrix} \end{matrix}$$

$$\text{Det } A = \begin{vmatrix} 3 & 2 & -5 \\ 2 & -1 & -2 \\ 1 & -2 & -3 \end{vmatrix} \begin{matrix} 3 \cdot 2 \cdot (-3) + 2 \cdot (-1) \cdot (-2) - (20 + 12 - 12) \\ 52 - 20 = \end{matrix} \boxed{\text{Det } A = 32}$$

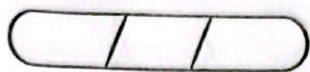
$$\text{Det } X = \begin{vmatrix} 8 & 2 & -5 \\ -1 & -1 & -2 \\ -1 & -2 & -3 \end{vmatrix} \begin{matrix} 8 \cdot 2 \cdot (-3) + 16 - 10 \\ -80 + 32 + 24 = \end{matrix} \boxed{92 + 24 = 96}$$

$$\text{Det } Y = \begin{vmatrix} 3 & 8 & -5 \\ 2 & -1 & -2 \\ 1 & -1 & -3 \end{vmatrix} \begin{matrix} L_1 \leftarrow L_1 + 2 \cdot L_2 \\ \rightarrow \begin{pmatrix} 7 & 0 & -9 \\ 2 & -1 & -2 \\ 1 & -1 & -3 \end{pmatrix} \\ L_2 \leftarrow L_2 + (-1) \cdot L_3 \rightarrow \begin{pmatrix} 7 & 0 & -9 \\ 1 & 0 & 1 \\ 1 & -1 & -3 \end{pmatrix} \\ -1 \cdot 1 \cdot 16 = \end{matrix} \boxed{64}$$

$$\text{Det } Z = \begin{vmatrix} 3 & 2 & 8 \\ 2 & -1 & -1 \\ 1 & -2 & -1 \end{vmatrix} \begin{matrix} L_1 \leftarrow L_1 + L_3 \\ \rightarrow \begin{pmatrix} 4 & 0 & 7 \\ 2 & -1 & -1 \\ 1 & -2 & -1 \end{pmatrix} \\ L_2 \leftarrow L_2 + 2 \cdot L_3 \rightarrow \begin{pmatrix} 4 & 0 & 7 \\ 0 & 0 & 1 \\ 1 & -2 & -1 \end{pmatrix} \\ 21 - 1 \cdot 8 = \end{matrix} \boxed{32}$$

$$X = \frac{\text{det } X}{\text{det } A} \rightarrow \frac{96}{32} \rightarrow 3 \quad \left| \quad Y = \frac{\text{det } Y}{\text{det } A} \rightarrow \frac{64}{32} \rightarrow 2 \right.$$

$$Z = \frac{\text{det } Z}{\text{det } A} \rightarrow Z = \frac{32}{32} \rightarrow \boxed{Z = 1}$$



$$e) \begin{cases} x + 2y - z = 2 \\ 2x - y + 3z = 9 \\ 3x + 3y - 2z = 3 \end{cases} \rightarrow \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \\ 3 & 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 9 \\ 3 \end{pmatrix}$$

$$\text{Det A} \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \\ 3 & 3 & -2 \end{pmatrix} \begin{matrix} 1 & 2 & (1 \cdot 2 + 18 - 6) \rightarrow 14 \\ 2 & -1 & -(3 + 9 + 8) \rightarrow -20 \\ 3 & 3 & -10 \end{matrix}$$

$$\text{Det X} \begin{pmatrix} 2 & 2 & -1 \\ 9 & -1 & 3 \\ 3 & 3 & -2 \end{pmatrix} \begin{matrix} 2 & 2 & (14 + 18 - 27) \rightarrow -5 \\ 9 & -1 & -(73 + 18 + 36) \rightarrow -127 \\ 3 & 3 & 0 \end{matrix} \rightarrow \boxed{+10}$$

$$\text{Det Y} \begin{pmatrix} 1 & 2 & -1 \\ 2 & 9 & 3 \\ 3 & 3 & -2 \end{pmatrix} \begin{matrix} 1 & 2 & -18 + 18 - 6 \rightarrow -6 \\ 2 & 9 & -(27 + 9 - 8) \rightarrow -28 \\ 3 & 3 & 0 \end{matrix} \rightarrow \boxed{+20}$$

$$\text{Det Z} \begin{pmatrix} 1 & 2 & 2 \\ 2 & -1 & 9 \\ 3 & 3 & 3 \end{pmatrix} \begin{matrix} 1 & 2 & -3 + 54 + 12 \rightarrow 63 \\ 2 & -1 & -(-6 + 27 + 12) \rightarrow -33 \\ 3 & 3 & 0 \end{matrix} \rightarrow \boxed{+30}$$

$$X = \frac{\text{det X}}{\text{det A}} \rightarrow X = \frac{10}{+10} \rightarrow \boxed{+1}$$

$$Y = \frac{\text{det Y}}{\text{det A}} \rightarrow Y = \frac{20}{+10} \rightarrow \boxed{+2}$$

$$Z = \frac{\text{det Z}}{\text{det A}} \rightarrow Z = \frac{30}{+10} \rightarrow \boxed{3}$$

F)

$$\begin{cases} x + 0 + z = -8 \\ 2x - 4y + 0 = -4 \\ 3x - 2y - 5z = 26 \end{cases} \quad \begin{pmatrix} 1 & 0 & 1 \\ 2 & -4 & 0 \\ 3 & -2 & -5 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -8 \\ -4 \\ 26 \end{pmatrix}$$

$$\text{Det A} = \begin{vmatrix} 1 & 0 & 1 \\ 2 & -4 & 0 \\ 3 & -2 & -5 \end{vmatrix} \begin{matrix} 10 \\ 20+0-4 \\ 3-2 \end{matrix} \begin{matrix} 10 \\ 16 \\ 3-2 \end{matrix} \rightarrow \begin{matrix} 10 \\ -12+0+10 \\ 1 \end{matrix} \rightarrow \boxed{28}$$

$$\text{Det X} = \begin{vmatrix} -8 & 0 & 1 \\ -4 & -4 & 0 \\ 26 & -2 & -5 \end{vmatrix} \begin{matrix} -10-0-5 \\ 10 \\ 10 \end{matrix} \rightarrow \begin{vmatrix} -8 & 0 & 1 \\ -4 & -4 & 0 \\ -18 & -2 & 0 \end{vmatrix} \begin{matrix} 1 \cdot (-1)^{1+3} \cdot (-4 \cdot -4) \\ 8+56 \\ 1 \cdot 1 \cdot 1 \cdot -18 \end{matrix} \rightarrow \boxed{-218}$$

$$\text{Det Y} = \begin{vmatrix} 1 & -8 & 1 \\ 2 & -4 & 0 \\ 3 & 26 & -5 \end{vmatrix} \begin{matrix} 15 \\ 10 \\ 10 \end{matrix} \rightarrow \begin{vmatrix} 1 & -8 & 1 \\ 2 & -4 & 0 \\ 8 & -14 & 0 \end{vmatrix} \begin{matrix} 1 \cdot (-1)^{1+3} \cdot (2 \cdot -4) \\ 8-14 \\ 1 \cdot 1 \cdot 1 \cdot 21 \end{matrix} \rightarrow \boxed{21}$$

$$\text{Det Z} = \begin{vmatrix} 1 & 0 & -8 \\ 2 & -4 & -4 \\ 3 & -2 & 26 \end{vmatrix} \begin{matrix} 10 \\ 10 \\ -6 \end{matrix} \rightarrow \begin{vmatrix} 1 & 0 & -8 \\ -4 & 0 & -56 \\ -3 & -2 & -24 \end{vmatrix} \begin{matrix} 1 \cdot (-1)^{1+3} \cdot (-2 \cdot -56) \\ -2 \cdot -56 \\ -2 \cdot -1 \cdot -88 \end{matrix} \rightarrow \boxed{-176}$$

$$X = \frac{\text{det X}}{\text{det A}} \rightarrow \frac{-218}{28} \rightarrow \frac{-24}{14} \rightarrow \boxed{\frac{-12}{7}}$$

$$Y = \frac{\text{det Y}}{\text{det A}} \rightarrow \frac{21}{28} \rightarrow \boxed{\frac{1}{7}}$$

$$Z = \frac{\text{det Z}}{\text{det A}} \rightarrow \frac{-176}{28} \rightarrow \frac{-88}{14} \rightarrow \boxed{\frac{24}{7}}$$

$$G) \begin{cases} X + 2Y + 3Z = 10 \\ 3X + 4Y + 6Z = 23 \\ 3X + 2Y + 3Z = 10 \end{cases} \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 6 \\ 3 & 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 10 \\ 23 \\ 10 \end{pmatrix}$$

$$\text{Det } A = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 4 & 6 \\ 3 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} \rightarrow 2 \cdot (-1)^{1+1} \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 2 \cdot (-2) = -4$$

$$\rightarrow \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 0$$

$$\text{Det } X = \begin{vmatrix} 10 & 2 & 3 \\ 23 & 4 & 6 \\ 10 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 10 & 2 & 3 \\ 23 & 4 & 6 \\ 0 & 0 & 0 \end{vmatrix} \rightarrow \text{Det} = 0$$

$$\text{Det } Y = \begin{vmatrix} 1 & 10 & 3 \\ 3 & 23 & 6 \\ 3 & 10 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 10 & 3 \\ 3 & 23 & 6 \\ 2 & 0 & 0 \end{vmatrix} \rightarrow 2 \cdot (-1)^{3+1} \begin{vmatrix} 10 & 3 \\ 23 & 6 \end{vmatrix} = 2 \cdot (-1) \cdot (60 - 69) = 9$$

Sistema impossível

$$5) a) \begin{cases} 3x - 2y = 0 \\ -6x + 8y = 0 \end{cases} \rightarrow \begin{pmatrix} 3 & -2 \\ -6 & 8 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \left| \begin{array}{cc|c} x & y & 0 \\ 1 & 0 & -2 \\ 0 & 1 & 2 \end{array} \right|$$

$$\det A = \begin{vmatrix} 3 & -2 \\ -6 & 8 \end{vmatrix} \rightarrow 3 \cdot 8 - (-2 \cdot -6) = 24 - 24 = 0$$

$$\det X = \begin{vmatrix} 0 & -2 \\ 0 & 8 \end{vmatrix} \rightarrow 0 \cdot 8 - (-2 \cdot 0) = 0$$

$$\det Y = \begin{vmatrix} 3 & 0 \\ -6 & 0 \end{vmatrix} \rightarrow 3 \cdot 0 - 0 \cdot -6 = 0$$

SPI $\rightarrow$ Sistema Possível Indeterminado

$$b) \begin{cases} x + y + z = 0 \\ 2x + 2y + 2z = 0 \\ x + y + z = 0 \end{cases} \quad \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\det A = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{vmatrix} \quad \det = 0 \text{ pela propriedade de que duas linhas/colunas iguais é igual a } \boxed{\det 0}$$

$$\det X = \begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{vmatrix} \quad \det = 0 \text{ pela propriedade de que se uma linha/ou coluna é a outra } \boxed{\det \text{ é } 0}$$

$$\det Y = \begin{vmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & 0 & 1 \end{vmatrix} \quad \det = 0 \text{ pois possui uma coluna toda de zeros.}$$

$$\det Z = \begin{vmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \\ 1 & 1 & 0 \end{vmatrix} \quad \det = 0 \text{ pois tem 2 colunas iguais}$$

SPI $\rightarrow$ Sistema Possível Indeterminado

$$c) \begin{cases} X + Y + 2Z = 0 \\ X - Y - 3Z = 0 \\ X + 4Y + 0 = 0 \end{cases} \quad \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & -3 \\ 1 & 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Det A

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & -3 \\ 1 & 4 & 0 \end{pmatrix} \begin{matrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 4 & 0 \end{matrix} \rightarrow \begin{matrix} 0+3+8=11 \\ (-2+12+0)=10 \end{matrix} \rightarrow \begin{matrix} 11 \\ 10 \end{matrix} \rightarrow \boxed{1}$$

Det X

$$\begin{pmatrix} 0 & -1 & 2 \\ 0 & -1 & -3 \\ 0 & 4 & 0 \end{pmatrix} \begin{matrix} 0 & -1 \\ 0 & -1 \\ 0 & 4 \end{matrix} \begin{matrix} 0+0 \\ 0+0 \\ 0+0 \end{matrix}$$

$$\boxed{\text{Det} = 0}$$

Det Y

$$\begin{pmatrix} 1 & 0 & 2 \\ 1 & 0 & 3 \\ 1 & 0 & 0 \end{pmatrix} \begin{matrix} 10 \\ 10 \\ 10 \end{matrix} \begin{matrix} 0+0 \\ 0+0 \\ 0+0 \end{matrix}$$

$$\boxed{\text{Det} = 0}$$

Det Z

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \\ 1 & 4 & 0 \end{pmatrix} \begin{matrix} 1 & -1 \\ 1 & -1 \\ 1 & 4 \end{matrix} \begin{matrix} 0+0+0 \\ 0+0+0 \\ 0+0+0 \end{matrix}$$

$$\boxed{\text{Det} = 0}$$

5 PD - Sistema Possível Determinado

$$(6) a) \begin{cases} 3x + my = 2 \\ x - y = 1 \end{cases} \rightarrow \begin{pmatrix} 3 & m \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{cases} 3 - m = 0 \rightarrow x = m \\ -2 - m = 0 \rightarrow y = -2 - m \\ 1 = 0 \rightarrow y = -x \end{cases}$$

Det A

$$\begin{pmatrix} 3 & m \\ 1 & -1 \end{pmatrix} \rightarrow 3 - m \rightarrow \text{Det A} = 3 - m$$

Det X

$$\begin{pmatrix} 2 & m \\ 1 & -1 \end{pmatrix} \rightarrow 2 - m \rightarrow \text{Det X} = 2 - m$$

Det Y

$$\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \rightarrow 3 - 2 = 1 \rightarrow \text{Det Y} = 1$$

SPD só é possível quando $m \neq 3$.

b)

$$\begin{cases} 3x + 2(m-1)y = 1 \\ mx - 4y = 0 \end{cases} \rightarrow \begin{pmatrix} 3 & 2m-2 \\ m & -4 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$-12 - (2m-2 \cdot m) \rightarrow -12 - (2m^2 - 2m) \rightarrow -2m^2 + 2m - 12 \div 2 \rightarrow -m^2 + m - 6$$

$$-b \pm \sqrt{b^2 - 4ac} \rightarrow -1 \pm \sqrt{1 - 24} \rightarrow -1 \pm \sqrt{-23}$$

Não é possível encontrar um valor para m .

$$c) \begin{cases} x - y = 2 \\ x + my + z = 0 \\ -x + y - z = 2 \end{cases}$$

$$\begin{cases} x = 2 \\ y = 0 \\ z = 2 \end{cases}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & m & 1 \\ -1 & 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 & | & 1 \\ 1 & m & 1 & | & m \\ -1 & 1 & -1 & | & 1 \end{pmatrix} \rightarrow \begin{matrix} -m+1+0 \rightarrow -m+1-2 \\ -0+1+1 \rightarrow -m-1 \neq 0 \end{matrix}$$

$$\rightarrow -m-1 \neq 0 \rightarrow m \neq -1$$

$$-m-1 \neq 0 \rightarrow m \neq -1$$

$m \neq -1$ Para ser um sistema possível determinado

$$d) \begin{cases} mx + y - z = 1 \\ x + my + z = 0 \\ x - y + 0 = 2 \end{cases} \rightarrow \begin{pmatrix} m & 1 & -1 \\ 1 & m & 1 \\ 1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

$$\text{Det } A \begin{pmatrix} m & 1 & -1 \\ 1 & m & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} m & 1 \\ 1 & m \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0+1+1 \\ -(m-m+0) \end{pmatrix} \begin{pmatrix} 2+2m \neq 0 \\ 2m = -2 \\ m \neq -1 \end{pmatrix}$$

7

$$\begin{aligned} 6x - 2y &= 750 \\ 1350 - 4y + x + y &= 225 \\ x &= 225 - y \\ 6(225 - y) - 2y &= 750 \\ 1350 - 6y - 2y &= 750 \\ -8y &= -600 \\ y &= 75 \end{aligned}$$

Se $y = 75$ $x = 150$

8

$$\begin{aligned} T + S &= 540 \\ 6T + 2S &= 3000 \\ T &= 540 - S \\ 6(540 - S) + 2S &= 3000 \\ 3240 - 6S + 2S &= 3000 \\ -4S &= -240 \\ S &= 60 \end{aligned}$$

$T + 60 = 540 \rightarrow T = 480$

9

$$\begin{aligned} 2x + 5y + 10z &= 500 \\ x + y + z &= 92 \\ 2x + 5y + 10z &= 500 \\ x + y + z &= 92 \end{aligned}$$

Logo $x = 20$ $z = 20$
 $y = 52$

10) $K + A = 109$

$$K + T = 172$$

$$T + A = 97$$

$$\rightarrow T = 97 - A$$

$$K + 97 - A = 172$$

$$K - A = 75$$

$$K + A = 109$$

$$K - A = 75$$

$$2K = 184$$

$$K = 92$$

$$77 + A = 109 \rightarrow$$

$$A = 109 - 77$$

$$A = 32$$

$$T = 97 - 32 \rightarrow$$

$$T = 65$$